Final Project Report: Predicting Income Category Using the Adult Dataset

Abstract

Predicting whether an individual's annual income surpasses fifty thousand dollars presents a meaningful challenge for researchers and practitioners working at the intersection of economics, social science, and machine learning. The task requires not only the development of accurate predictive models but also a nuanced understanding of the structural factors that influence wages and economic mobility. This project constructs a supervised classification system that draws upon demographic, educational, and employment-related attributes to infer income categories. The analysis utilizes a dataset containing more than forty-eight thousand individual records and implements a complete data mining workflow, including data cleaning, exploratory assessment, feature engineering, feature selection, model construction, hyperparameter optimization, and evaluation. Several machine learning algorithms were explored, with Decision Trees, Naive Bayes, and K-Nearest Neighbors forming the core of the comparison. Extensive preprocessing addressed missing values, outliers, redundant variables, and categorical inconsistencies, all of which are common in large-scale socioeconomic datasets. The tuned Decision Tree ultimately emerged as the strongest performer, achieving an accuracy of 83.4 percent. Additional analysis revealed that education level, marital status, age, hours worked per week, and occupation were among the most influential predictors of higher income. While the resulting models demonstrate strong discriminatory power, the project acknowledges limitations such as class imbalance, absence of geographic and industry-specific variables, and constraints of simpler ensemble approaches. Expanding the modeling framework through richer features, more advanced ensemble methods, and synthetic data balancing represents a promising direction for future enhancements.

Introduction

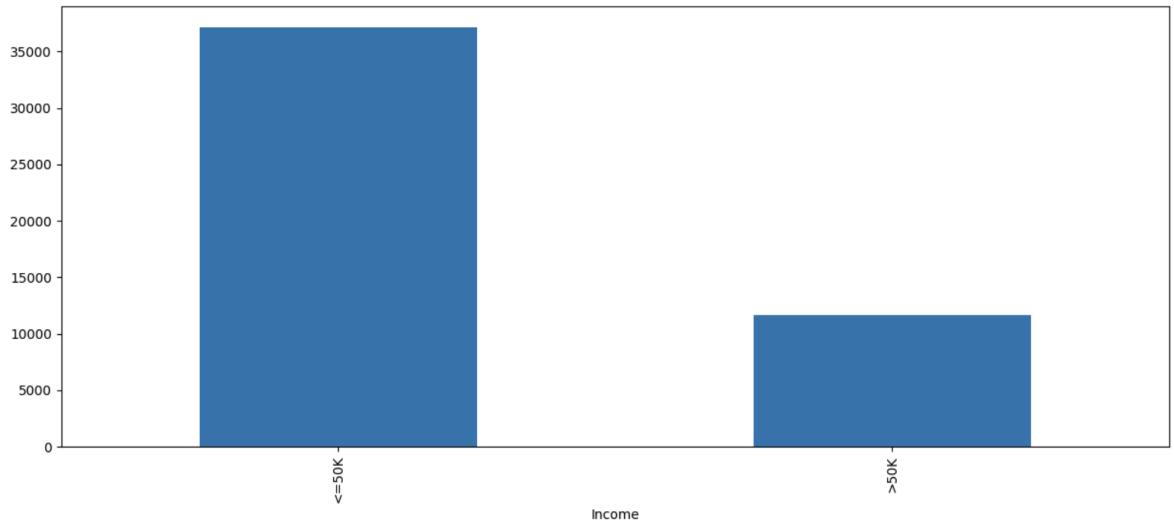
The problem of estimating an individual's income level based on demographic and employment characteristics has long been of interest across numerous academic and professional fields. Economists use income prediction models to study wage inequality and socioeconomic mobility. Human resource specialists rely on such models to support compensation benchmarking and workforce planning. Public policy researchers investigate the role of demographic attributes in shaping economic outcomes, and financial institutions leverage predictive income assessments for credit scoring and risk evaluation. Across these domains, machine learning offers an efficient and powerful means of uncovering complex, nonlinear relationships within diverse socioeconomic data.

This project focuses on predicting whether an individual earns more or less than fifty thousand dollars per year. The dataset contains nearly forty-nine thousand observations and includes a broad range of demographic and employment variables such as age, sex, race, education, marital status, occupation, workclass, capital gains, capital losses, and the number of hours worked per week. The target variable divides individuals into two distinct income categories. A key challenge arises from the distribution of these categories: only approximately one-quarter of individuals belong to the higher-income group, creating a notable class imbalance. This imbalance has direct implications for training machine learning models, as many algorithms may gravitate toward favoring the majority class unless specific corrective strategies are employed.

The primary goal of this project is to develop a predictive model that not only achieves high classification accuracy but also provides a meaningful understanding of the socioeconomic factors most strongly associated with higher income levels. To accomplish this, the study utilizes

a structured data mining pipeline, beginning with data preparation and exploratory analysis, followed by feature engineering and feature selection, and concluding with model training, hyperparameter tuning, and performance evaluation. This approach helps ensure that the final model is both accurate and interpretable, offering insights into the dynamics of income prediction.

Figure 1 — "Income Category Distribution" (Box Plot)



Experiments

Table 1 — "Summary of Dataset Features"

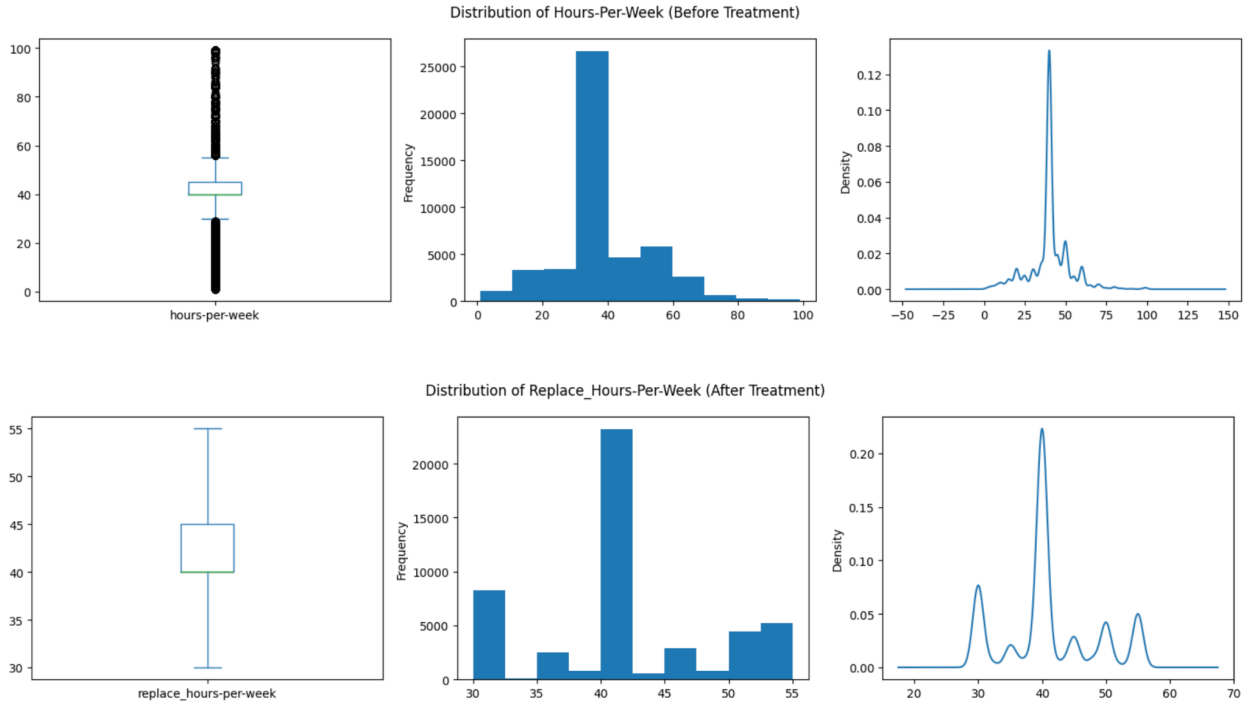
Attribute	*Represents*	*Type*	*Notes / Missing / Usefulness*
age	Age of individual	Numeric	No missing; likely predictive
workclass	Employment type (Private, Govt, etc.)	Categorical	?' indicates missing; useful for income prediction
fnlwgt	Census weighting factor	Numeric	Not directly predictive; usually dropped
education	Education level	Categorical	Strongly predictive
education-num	Education level numeric	Numeric	Redundant with education; choose one
martial-status	Marital status	Categoricial	Likely predictive
occupation	Job type	Categoricial	Missing values present; highly predictive
relationship	Role in household	Categoricial	Often predictive
race	Race identity	Categoricial	Requires fairness consideration
sex	Gender	Categoricial	Requires fairness consideration
captial-gain	Capital gain earnings	Numeric	Skewed; strong predictor
capital-loss	Capital loss	Numeric	Skewed; strong predictor
hours-per-week	Weekly working hours	Numeric	Predictive
native-country	Country of origin	Categorical	Many categories; may be grouped
income	Target (>50K or ≤ 50K)	Categorical	Classification label

Dataset Characteristics. The dataset used for this study consists of more than forty-eight thousand records and includes fifteen primary attributes describing demographic and employment-related characteristics. Numerical attributes include age, hours worked per week, capital gain, and capital loss. Categorical attributes cover variables such as occupation, marital status, workclass, race, and educational background. Together, these attributes represent a diverse set of socioeconomic indicators that can influence a person's earning power. The target variable identifies whether an individual earns more than fifty thousand dollars annually. Because only about twenty-four percent of individuals fall into the higher-income category, the dataset exhibits a substantial class imbalance that must be accounted for throughout the modeling process.

Data Cleaning. Initial data exploration revealed several data quality issues that required careful handling. Some variables, particularly workclass, occupation, and native-country, contained missing or ambiguous values. These were addressed through strategic imputation and category consolidation to prevent model bias and data fragmentation. Outliers were another significant concern. Using an interquartile-range–based method, extreme values were identified in

numerical variables such as age, capital-gain, capital-loss, and hours-per-week. To ensure that these outliers did not distort model performance, they were capped or replaced with statistically justified boundary values. For example, implausibly high or low age values were corrected, and the hours-per-week variable was constrained to a realistic working range between approximately 30 and 55 hours.

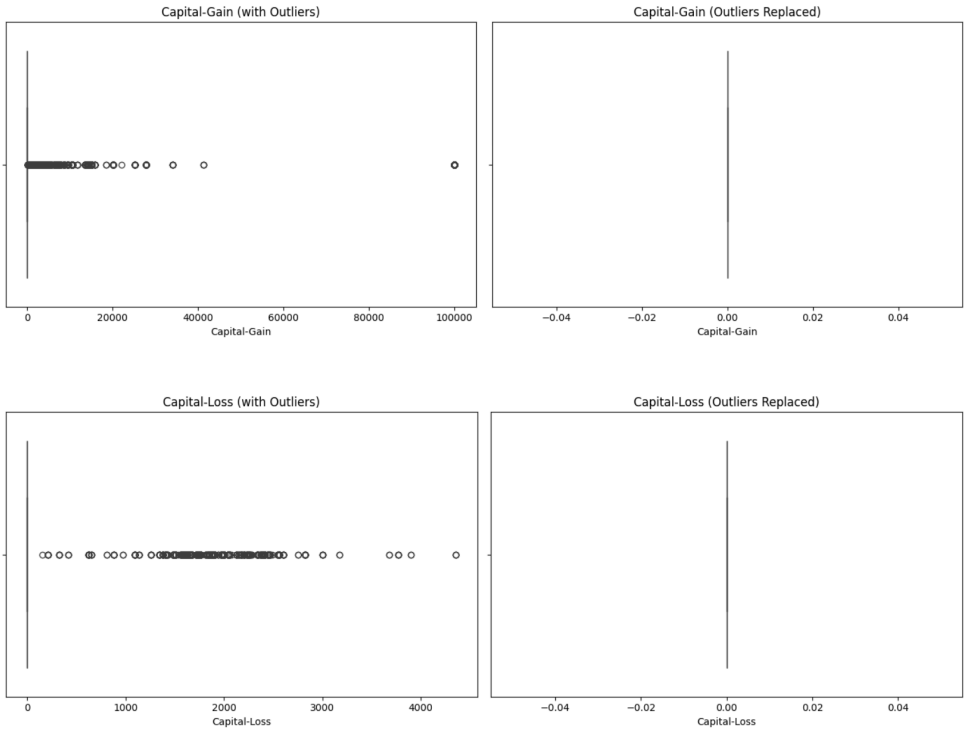
Figure 2 — Histogram of Age (Before and After Cleaning)



Other variables displayed heavily skewed distributions, particularly capital-gain and capital-loss, which are near-zero for most individuals but extremely high for a few. Rather than removing these features, the project reduced their extreme values to improve stability while preserving meaningful information. Redundancy was also addressed: the dataset included both an education label and a corresponding numerical code, but because the numerical version sufficiently represented educational attainment, the redundant label was removed. Finally,

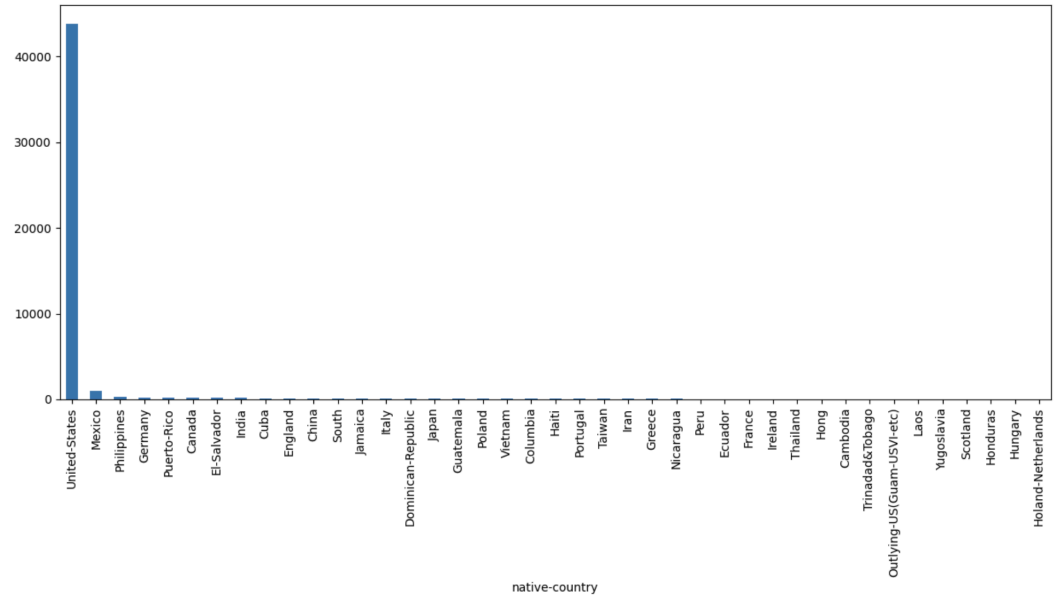
categorical values were standardized, ensuring that inconsistent formatting did not artificially inflate the number of categories.

Figure 3 — Boxplots of Capital-Gain and Capital-Loss



Processing Categorical Variables. Many categorical features required significant structural refinement. Some categories appeared extremely infrequently, especially in attributes like native-country. These rare categories were consolidated into an “Other” classification to reduce dimensionality and prevent sparsity during encoding. Textual inconsistencies like minor formatting differences or spacing issues were corrected to maintain a uniform representation of categories. After this standardization, all categorical features were prepared for one-hot encoding, resulting in a feature set well suited for algorithms that operate in numerical space. This transformation expanded the dataset into a higher-dimensional representation but ensured that each categorical distinction was preserved in a model-compatible format.

Figure 4 — Bar Chart of Native-Country Frequencies



Feature Engineering. Feature engineering contributed meaningfully to the overall predictive performance. Age, for example, was transformed into bins representing distinct life stages: young adulthood, mid-career, late career, and retirement age. These bins capture nonlinear earning patterns that a raw numeric age variable might not fully express. Additional binary indicators were created to highlight key economic behaviors, such as whether an individual had any capital gains or losses and whether they worked unusually long hours. These binary features help differentiate individuals with distinct financial or employment profiles that correlate strongly with income. Continuous variables were standardized to improve model stability, particularly for algorithms sensitive to scale differences. With these engineered transformations, the dataset became more expressive and capable of capturing subtle predictive relationships.

Figure 5 — Age Binning Diagram

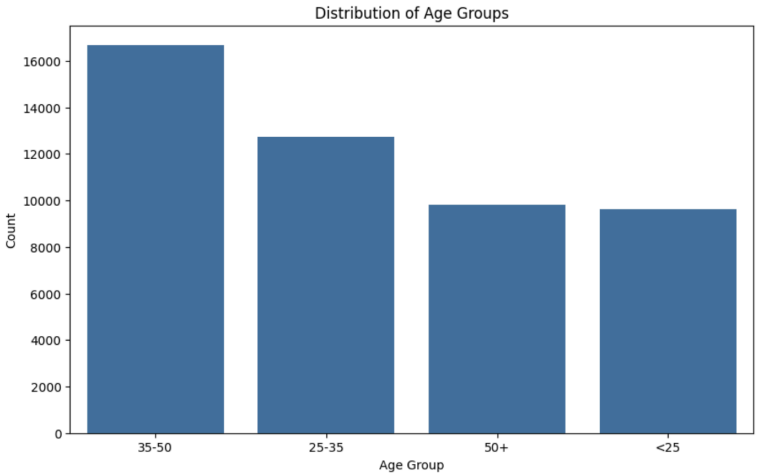
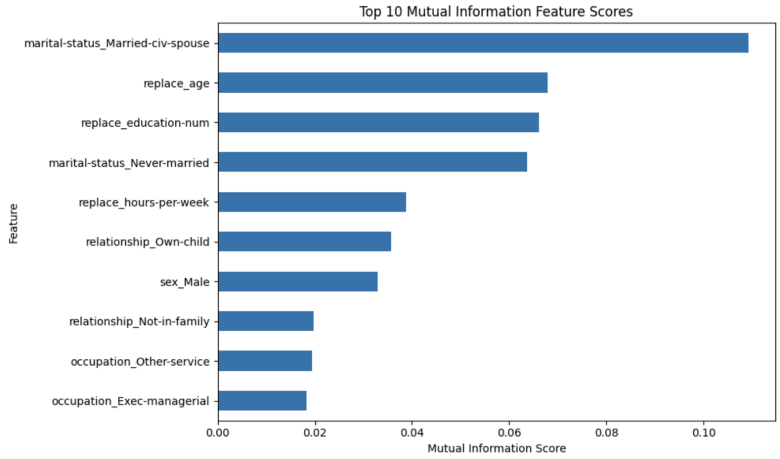


Table 2 — "Engineered Feature Definitions"

Engineered Feature	***Type***	***Description***
has_gain	Binary	Indicates capital-gain > 0
has_loss	Binary	Indicates capital-loss > 0
works_long_hours	Binary	Hours per week > 50
age_group	Categorical	Age category assignment

Feature Selection. To improve model efficiency, reduce noise, and enhance interpretability, the project applied mutual-information–based feature selection. This technique measures how much information each feature contributes to predicting the target variable. Several attributes emerged as particularly influential: education level, age, hours worked per week, marital status, and occupation. These results align with long-established socioeconomic theories that connect income to education, family structure, work commitment, and job type. From over one hundred possible encoded features, a subset of about twenty-one was selected for the final modeling stage. Reducing the dimensionality not only accelerated training but also improved model consistency and generalization by removing irrelevant or weakly informative attributes.

Figure 6 — Mutual Information Feature Scores (Bar Chart)



Model Development and Hyperparameter Tuning. Three supervised machine learning models served as the foundation for this analysis: Naive Bayes, K-Nearest Neighbors (KNN), and Decision Trees. These models were chosen for their interpretability, contrasting assumptions, and suitability for mixed-type datasets.

Baseline testing showed clear differences in their performance. KNN performed the strongest among the baselines, with an accuracy of approximately 82%. The Decision Tree achieved around 79% accuracy, while Naive Bayes underperformed at roughly 59%. To improve performance, extensive hyperparameter tuning was conducted using grid search. For KNN, the optimal settings included using eleven neighbors, Euclidean distance, and uniform neighbor weighting. For the Decision Tree, the best configuration included a maximum depth of nine, the entropy criterion for splitting, and a minimum of ten samples per leaf node. These parameters balanced model complexity with generalization ability, preventing both underfitting and overfitting.

An ensemble model based on soft voting was also implemented, combining predictions from multiple algorithms. However, the ensemble did not improve upon the tuned Decision Tree because the base models were not sufficiently diverse, and the relatively weak Naive Bayes model diluted the combined performance.

Results

Table 3 — "Baseline Model Metrics"

Model	*Accuracy*	*Notes*
KNN	~0.82	Strongest baseline
Decision Tree	~0.79	Good interpretability
Naive Bayes	~0.59	Weak due to feature correlations

The results of the modeling process demonstrate meaningful performance differences among the algorithms tested. The strongest overall performer was the tuned Decision Tree, which achieved an accuracy of 83.4%. The tuned KNN model followed closely with an accuracy of 82.8%. Both of these models exhibited significant improvements over their baseline versions, confirming the value of careful hyperparameter tuning. The confusion matrices for each model indicated that the Decision Tree maintained a balanced ratio of true positives to true negatives, making it especially well suited for handling the dataset's class imbalance.

Figure 7 — Confusion Matrix: Tuned Decision Tree

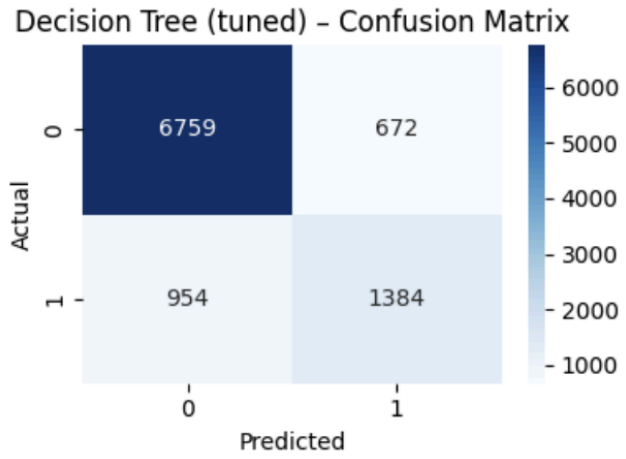
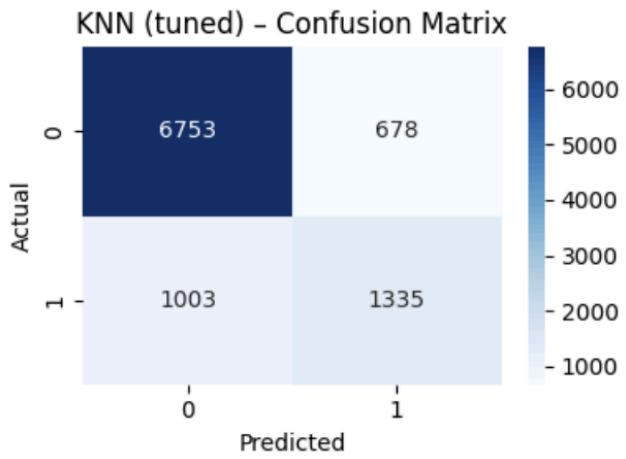


Figure 8 — Confusion Matrix: Tuned KNN



Naive Bayes, by contrast, continued to underperform even after tuning. The model's simplifying assumptions were unsuitable for a dataset in which variables such as education, occupation, and marital status are highly correlated. The ensemble experiment did not yield superior results either. Because the models shared similar weaknesses and did not contribute sufficiently distinct decision patterns, their combined output achieved a lower accuracy of around 80%.

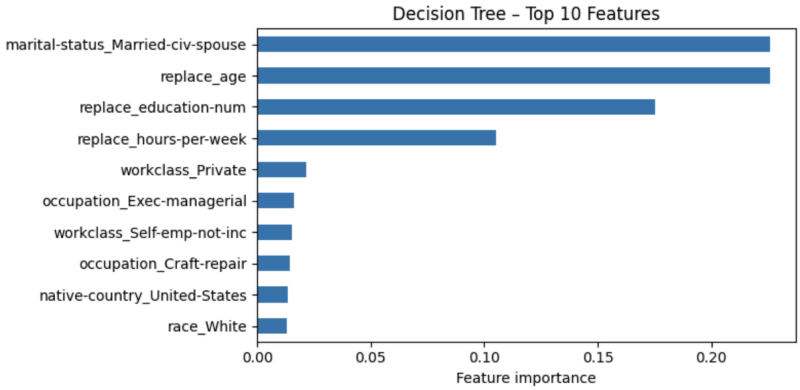
Table 4 — "Tuned Model Comparison"

Model	***Accuracy***
Decision Tree	0.834
KNN	0.828
Ensemble	~0.80

Feature importance analysis provided additional insight. Marital status emerged as the single most predictive attribute, followed by education level, age, hours worked, occupation, and relationship type. These findings support widely held economic perspectives: individuals with higher education levels, stable marital relationships, more work experience, and longer working hours tend to have greater earning potential. The consistency of these predictors across both

mutual information ranking and model-based feature importance analyses strengthens confidence in their relevance.

Figure 9 — Decision Tree Feature Importance Plot



Conclusion

This study successfully constructed a predictive model for determining whether an individual earns more than $50,000 annually based on a combination of demographic and employment characteristics. The project demonstrated the essential role of data cleaning and feature engineering in improving model performance. By addressing missing values, correcting outliers, reducing redundancy, and standardizing categorical variables, the dataset was transformed into a stable foundation for classification. Feature engineering and selection further refined the dataset, highlighting the socioeconomic patterns embedded within the data.

The Decision Tree model ultimately produced the strongest predictive performance, balancing accuracy, interpretability, and generalization. Consistent with established socioeconomic research, the model identified education, marital status, age, hours worked per week, and occupation as key determinants of income. Despite the model's strengths, certain limitations remain. The dataset's class imbalance may reduce performance on minority income classes, and the absence of geographic or industry-specific details limits the model's contextual

richness. Additionally, the unsuccessful ensemble highlights the importance of selecting diverse base learners when constructing combined models.

Future enhancements to this work may involve incorporating more advanced algorithms such as Random Forests, Gradient Boosting Machines, or XGBoost, all of which are adept at capturing nonlinear interactions and feature hierarchies. Synthetic resampling methods such as SMOTE may help mitigate class imbalance, while stratified k-fold cross-validation would yield a more robust measure of model performance. Further engineering of interaction features could also prove valuable in uncovering layered relationships between attributes.

Overall, this project illustrates how a structured data mining process can yield both accurate predictive models and deeper insights into the socioeconomic variables that shape income outcomes. The results underscore the importance of careful preprocessing, thoughtful feature design, and systematic model evaluation in transforming raw data into meaningful, interpretable knowledge.